Octav I. Dragoi

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EDUCATION

Master of Science 2018 - 2020
Technische Universität München, Munich, Germany
Program: Data Engineering and Analytics.
Grade: 1.385. (1 to 5, 1 is best)

Coursework: Foundations of Data Engineering, Nonlinear Optimization: Advanced, Introduction to Deep Learning, Foundations of Data Analysis, Combinatorial Optimization.

Bachelor of Arts 2012 - 2016
Harvard College, Cambridge, MA
Concentration: Mathematics and Computer Science.
Secondary: Economics.
Overall GPA: 3.75/4.00

Coursework: Honors Abstract Algebra, Honors Real and Complex Analysis, Differential Topology, Analytic Number Theory, Algebraic Geometry; General Computer Science, Data Structures and Algorithms, Machine Learning, Data Science.

EXPERIENCE

Quantitative Researcher Fall 2020 - present
GQS - Citadel LLC, London, UK & Miami, FL
The mid-horizon quantitative trading branch of the multistrategy hedge fund.

- Developed mid and long horizon trading signals, as part of an alpha research team.
- Gained wide experience trading equities and handling a variety of conventional and alternative datasets across all world regions.

Algorithmic Trader Summer 2016 - Fall 2018
Jump Trading LLC, New York, NY
A quantitative, proprietary trading company and registered market maker.

- Part of a team of 8 people trading equities worldwide, studying statistical properties of market data and implementing black-box trading algorithms.
- Created a framework that improves the entire team's performance by 20%. Designed and researched several new trading strategies. Implemented several essential components and features for the trading platform.
- Working on the full process pipeline: feature design, data analysis, phenomenon modeling, algorithm implementation, production code maintenance and efficientization. All work done in a Linux environment.

Quantitative Summer Analyst Intern Summer 2015

D.E. Shaw & Co., New York, NY

A large hedge fund, well known for employing quantitative strategies.

- Studied trend filtering methods and models that remove short-term noise and retain only the long-term significant movements. Developed new methods and compared them with already existing ones.
- Implemented a Python package of the aforementioned theoretical models and researched its practical predictive power on timeseries of market data.

Summer Analyst Intern Summer 2014

Ellington Management Group LLC, Old Greenwich, CT

A Connecticut-based hedge fund focused on trading mortgage-based securities. Worked in two separate departments, statistical arbitrage and research.

- Focused on creating higher frequency stat-arb signals. Did statistical analysis and data mining on intraday order book data for index ETFs.
- Worked on an agent-based housing market model, meant to reproduce and explain the subprime mortgage crisis associated with the Great Recession.

AWARDS

Putnam Mathematical Competition

- 10th place in 2012, in the N1 tranche of the national rankings.
- Honorable Mentions in 2013 and 2014.

International Mathematical Olympiad

- Gold Medal in 2011, Amsterdam, the Netherlands.
- Silver Medal in 2012, Mar del Plata, Argentina.

Balkan Mathematical Olympiad

- Gold Medal in 2010, Chisinau, Moldova.
- Gold Medal in 2011, Iasi, Romania.
- Gold Medal in 2012, Antalya, Turkey.

OTHER ACTIVITIES

Olympiad Math Instructor at UpperSchool Bucharest 2019 - present

Olympiad Math Instructor at BIMSA Summer Camp in Beijing, China 2024

PUBLICATIONS

Klischat, Moritz, Octav Dragoi, Mostafa Eissa and Matthias Althoff. "Coupling SUMO with a Motion Planning Framework for Automated Vehicles." *SUMO* (2019).

COMPUTER & LANGUAGE SKILLS

Programming: Python, C++, R, bash, LaTeX

Languages: Romanian (native), English (proficient), Russian, German (conversational).